

Probe Impedance Check User Guide

Ken L., Verasonics July 9, 2021

Basic Functionality

This tool, named “ProbeImpedanceCheck”, is intended to provide the user with a very quick and simple way to measure the impedance of each element of a probe that is connected to the system. It is located in the “Tools “ folder, at “Tools / SimpleImpedance_Tool”. It’s intended purpose is to identify suspected shorts to ground (measured impedance less than about 10 Ohms) or open circuits (measured impedance greater than several hundred Ohms). For elements with a measured impedance between those two levels, it will provide a rough estimate of the impedance magnitude, with accuracy limited to approximately +/- 25%. The impedance is measured at a single frequency, as specified by the user. Two usage models are envisioned: the tool can be invoked by simply entering “ProbeImpedanceCheck” at the Matlab command prompt, or it can be called as a function so it can be used within some other Matlab function or script such as a user’s SetUp script.

Command Prompt Usage

When the tool is invoked at the Matlab Command prompt with no arguments, it will go through the steps listed below, to measure impedance at the probe center frequency as given by the Trans structure. Alternatively, the tool can be invoked from the command prompt with an argument specifying the frequency in MHz at which impedance is to be measured. In this case the same steps will be followed, but the specified frequency will be used instead of Trans.frequency:

1. A check will be made to see if a Vantage hardware system is present and is not in a fault state. If this is not the case, it will exit with an error message.
2. The probe connector(s) of the system will be queried to determine whether something is connected at one of the connectors, starting with connector 1. If something is connected and if it has an ID that can be read by the system and if that ID is recognized by computeTrans, the computeTrans data will be used to identify the probe center frequency, element count and element signal mapping, and HVMux presence. With these values the tool will automatically measure the impedance at each element and present the results to the user as described below.
3. If nothing is detected at the probe connectors, or if the UTA does not support connection sensing or probe ID checking, the tool will then look in the Matlab workspace for a Trans structure (such as would be present if the user had just run a SetUp script). If a Trans structure is found, the parameters it provides will be used to configure the impedance check and it will proceed automatically to make the measurements and present the results.
4. If no Trans structure data was found in previous steps 2 and 3, the tool will proceed with automatically measuring the impedance at connector 1 for all element signals supported by that connector and the system configuration, at a default frequency of 5 MHz.

Presenting the Results

After the tool has finished the measurements, it will create a plot showing the measured impedance at each element with a heading that identifies the system configuration, probe name, and frequency used for the measurements. Any elements that appear to be open or shorted to ground will be identified and listed. It will also leave a structure “ProbeZdata” in the Matlab workspace, containing the measurement results and conditions used for the measurements. Note that since this tool invokes VSX to acquire the data and VSX runs in the base workspace, any variables the user had left in the workspace will be cleared.

Optional Arguments

When invoking “ProbImpedanceCheck” at the Matlab command prompt, up to three optional arguments can be provided, defined as follows:

1. **Measurement frequency:** the first argument must be a numeric value, specifying the frequency in MHz at which the measurement is to be made. If this value is specified it will override the default value that would have been taken from the Trans structure. However, if a Trans structure is to be provided as a second argument and you want to use the frequency from that structure, you can enter a value of zero for the first argument to indicate that it should be ignored.
2. **Trans Structure:** the second argument must be a Matlab structure, containing the following fields:
 - numelements: the number of elements in the probe
 - ConnectorES: the mapping of elements to element signals at the probe connector. If this field is not present, a default 1:1 mapping will be used.
 - frequency: the nominal center frequency of the probe. This value will be used for making the impedance measurements, unless a non-zero value was provided in argument 1.
 - HVMux: if the probe uses HVMux switches, a full HVMux structure must be provided to allow programming the mux devices in the probe.

Note that all of the other fields that are typically present in a Trans structure are not required, and will be ignored if they are present.

3. **Connector Selection:** for a UTA with more than one connector, this argument can be used to identify which connector (or set of connectors) are to be included in the measurements. When this field is not present the tool will default to connector 1.

Calling as a Function

Alternatively, “ProbImpedanceCheck” can be called as a function with an output argument, either at the Matlab command prompt or from within another script or function. When this is done the optional input arguments behave in the same way as described above, but the output impedance plot will not be generated and nothing will be displayed at the Matlab command prompt. The structure “ProbeZData” containing the system configuration and measurement results will be returned as the output argument. For a detailed listing of the input and output arguments, enter “help ProbImpedanceCheck” at the Matlab command prompt.